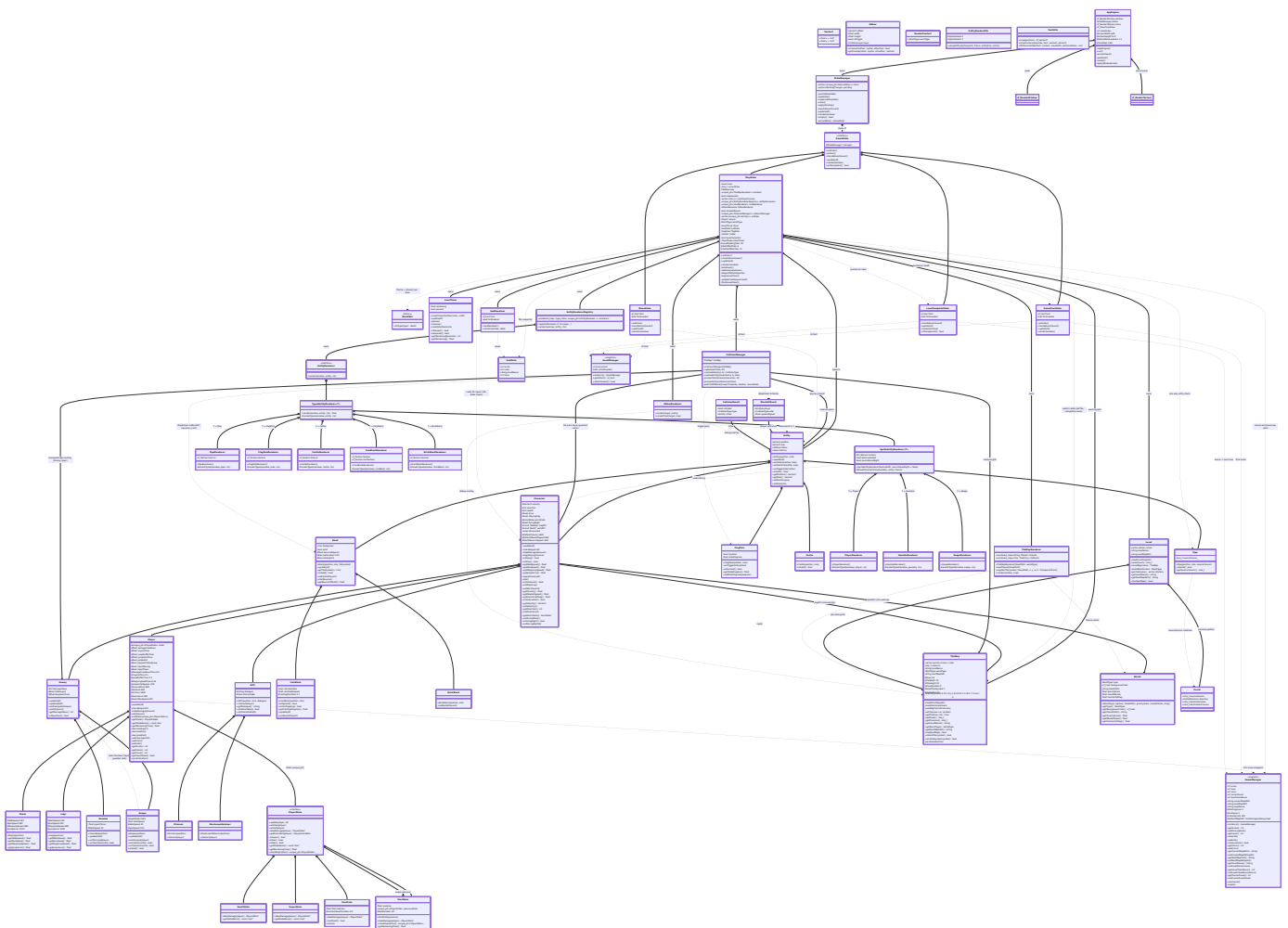


Full Project Class Diagram

Mermaid diagram covering all model, view, and controller classes (k.2.5). Stage 1 (model interface segregation) and stage 2, subtasks 0–3 are reflected: the goal castle is painted into the TileMap completion zone from a 21-symbol sheet (`TileMap::CastleSymbols` , registered by `TileMapRenderer` from atlas rows $y = 696/712$ (tower) and $728/744/760$ (base)); pipes spawn from 'p' (body) / 'P' (mouth) runs in the grid; and tile collision treats castle symbols as ground/solid. Still pending: stage 2 subtasks 4–6 (`SpritePainter` unification, removing the `Castle` entity + `CastleRenderer`) and stage 3 (the `PlayState` split) — the temporary `Castle` entity and `CastleRenderer` remain in the diagram until then. Companion doc: `CLASS_ENCAPSULATION_AND_CONNECTIONS.md` (per-class encapsulation + inter-class methods).



Enums

Enum	Values
CollisionType	None, Top, Bottom, Left, Right
CollisionLayer	Player, Enemy, Environment, Trigger
AnimState	Idle, Walk, Run, Jump, Fall, Die
PlayerAnimState	SmallIdle, SmallWalk, SmallRun, SmallJump, SmallFall, SmallDie, BigIdle, BigWalk, BigRun, BigJump, BigFall, FireIdle, FireWalk, FireRun, FireJump, FireFall, FireShoot, StarIdle, StarWalk, StarRun, StarJump, StarFall
KoopaState	Walking, ShellIdle, ShellSpinning
PortalDirection	Down, Up
WorldType	Overworld, Underwater, Castle

Legend

Arrow	Meaning
< --	Inheritance
*--	Composition (owning)
o--	Aggregation (non-owning reference)
..>	Dependency / usage